CHEST PRESS BY ECM

MORE TENSION ON THE CHEST — LESS STRESS ON THE SHOULDER

When you press, your body is not a rigid lever. The **Expansion-Compression Model (ECM)** treats your rib cage, breathing and shoulder blades as a moving system: where your body is allowed to **expand** and where it **compresses** decides whether the tension actually lands in the chest — and not in the shoulder, triceps or connective tissue.

01 THE SETUP — THE "STACK"

× "CHEST UP, EXTREME ARCH"

The lower rib cage is flared open and the back is heavily overextended. The rib cage **loses its shape and volume** — and with it the real stretch on the middle and lower chest fibers.

✓ "STACK" THE TORSO LIKE A CYLINDER

The ribs stay down and the torso forms a **stable cylinder** with a slight, natural arch. Pressure inside the rib cage — also between the shoulder blades — is kept.

THE PLASTIC-BOTTLE PRINCIPLE

Your rib cage is like a plastic bottle and your chest muscles are the label. Bend or crush the bottle (an extreme arch) and the label goes **loose**. Only a full, stacked rib cage keeps the muscles **under tension across their whole surface**.

02 SHOULDER BLADES — FLOATING ANCHORS

× PINNING & PRE-STRETCHING

Shoulder blades jammed hard down and back the whole time, with the chest "pre-stretched" beforehand: the upper back locks up, tension shifts to the **triceps and front shoulder** — and the chest can never fully contract at the top.

✓ LOW, MOBILE, GOING ALONG

Pull the shoulder blades **slightly down** but keep them mobile — they follow the arm. As you press out, they glide **slightly forward** at the end. The stretch happens smoothly under load, not rigidly beforehand.

03 BREATHING – YOUR STRETCHING TOOL

Don't just drop passively into the stretch. **Breathe in through your nose in a controlled way as you lower** and inflate the rib cage from the inside out — to the front, to the sides *and* into the back.

> **DOUBLE EFFECT**
> The added volume stretches the chest fibers **from the inside** — and the expansion in the back creates the room for the head of the upper arm to **glide cleanly backward**. Without that room it gets pushed forward: the classic cause of **shoulder pain when pressing**.

04 EXECUTION – THE 90-DEGREE RULE

DEEPER IS NOT BETTER

Lower your elbows only to about **90 degrees** — the way bodybuilding legends like Ronnie Coleman already did. That range holds the maximum torque at the joint. Stopping here while the torso stays stacked creates **far more real muscle tension** than forcing the joints into a bad leverage position for two extra centimeters.

STOP AT ABOUT 90° AT THE ELBOW

> ! Go too deep and your body has to break the stack, buckling the spine & ribs to make room. Muscle tension then shifts into **passive connective tissue, the front shoulder capsule and the triceps** — exactly where you don't want it.

BY THE WAY: ARCHING TURNS A FLAT BENCH INTO A DECLINE

An extreme arch only changes the angle of the ribs, and with it the line of pull on the muscle fibers — biomechanically you're then pressing a **decline bench with a broken stack**. More honest and more effective: just use a **slightly negative (decline) bench**. It isolates the middle and lower chest ideally while your torso stays stable — and brings the shoulder blades down and forward on their own.

05 CUES — HOW TO FEEL YOUR CHEST

CUE 01 · THE MOST IMPORTANT

COUNTERFORCE: PUSH YOURSELF AWAY FROM THE WEIGHT

Don't just think "press the weight forward." Imagine you are pushing **yourself backward deep into the bench** while the weight travels forward. This stabilizes the joints, prevents cheating and channels the tension **very locally into the chest**.

CUE 02

HAND PRESSURE POINTS

Place the pressure deliberately on the **outer heel of the hand** and the area **between thumb and index finger**. This lines up the wrist and elbow ideally and creates the internal rotation that pressing needs.

CUE 03 · ON HEAVY SETS

DON'T JAM YOUR HEAD INTO THE BENCH

When it gets heavy, many people push their head into the bench and overextend the neck. Better: nudge the head **slightly forward**, as if making your neck long. This creates room in the upper back — the chest has to take the load locally instead of dumping it onto the spine.

CUE 04

PRESS ALTERNATING

Both arms at once = massive compression on the body. **One arm at a time, alternating** (machine, cable) lets one side expand while the other compresses — this spares the joints and allows millimeter-precise tension on the chest.

06 NOT EVERY RIB CAGE IS THE SAME

The angle of your rib flare (the infrasternal angle) decides which strategy suits you — your rib cage is the platform for your shoulder joint:

NARROW RIB FLARE

Naturally geared toward expansion — **compression comes hard**. Helpful: machines and grips that force more internal rotation, plus **firm, closed fists** to create more compression artificially.

WIDE RIB FLARE

Compresses the upper back and rib cage **very quickly**. Helpful: exercises and grips that open the rib cage slightly — e.g. **open hands or cuffs on flys** — so the shoulder blades don't lock up right away.

07 FLYS – STRETCH WITH SENSE

The trend of forcing a maximally extreme stretch on flys — dumbbells almost to the floor — sounds "scientific" but is a trap.

WHERE YOUR RANGE OF MOTION REALLY ENDS

As soon as you lower the dumbbells so far that your rib cage loses its shape, you're **no longer stretching the chest muscle** — you're just hanging passively in connective tissue and joint capsules. Your usable range ends exactly where you **can no longer hold the stack**. Stopping just short of the extreme overstretch produces more real muscle tension than the uncontrolled maximum stretch.

STRETCH PHASE

OPEN HAND / CUFFS

An open hand (e.g. with arm cuffs on the cable) means less compression through the hand — the body stays in an **open, expanded position**. Ideal for the stretch phase.

PEAK CONTRACTION

CLOSED FIST

In front of the chest, in the shortened phase, a **firm fist** is clearly better: gripping hard signals the nervous system to compress and contract maximally. That's why it pays to switch from cuffs to handles or dumbbells within the same workout.

× JUST "HUGGING A TREE"

Just bringing the hands together while leaving the shoulder blades **pinned back** ends the movement before the chest ever got to work fully.

✓ PUSH FORWARD AT THE END

When your hands meet in front of the chest: **push the shoulder blades forward** and actively press the arms away to the front. Only this activates the full density of the chest.

! REMEMBER

Stack the torso instead of arching, keep the shoulder blades low and mobile, **breathe in as you lower**, stop at about **90 degrees** — and push yourself away from the weight as you press. That keeps the tension in the chest, not in the shoulder and connective tissue.